

# Switched Mode Power Supply (SMPS) Project Report

## Abstract

This project presents the design and implementation of a Switched Mode Power Supply (SMPS) used to convert input voltage into a stable DC output with high efficiency.

## Objective

The main objective is to design a reliable and efficient power supply circuit and develop its PCB layout for practical applications in electronics and medical devices.

## Design Overview

The system is based on high-frequency switching techniques to reduce power losses and improve efficiency compared to linear regulators.

## Main Components

The design includes switching controller IC, transformer, diodes, capacitors, inductors, and feedback control elements.

## PCB Design

The PCB was designed to minimize noise, reduce losses, and ensure proper thermal management and safe current paths.

## Conclusion

This project helped in understanding power electronics, PCB design techniques, and practical implementation of SMPS systems.